NEAR-REAL-TIME COASTAL INTELLIGENCE

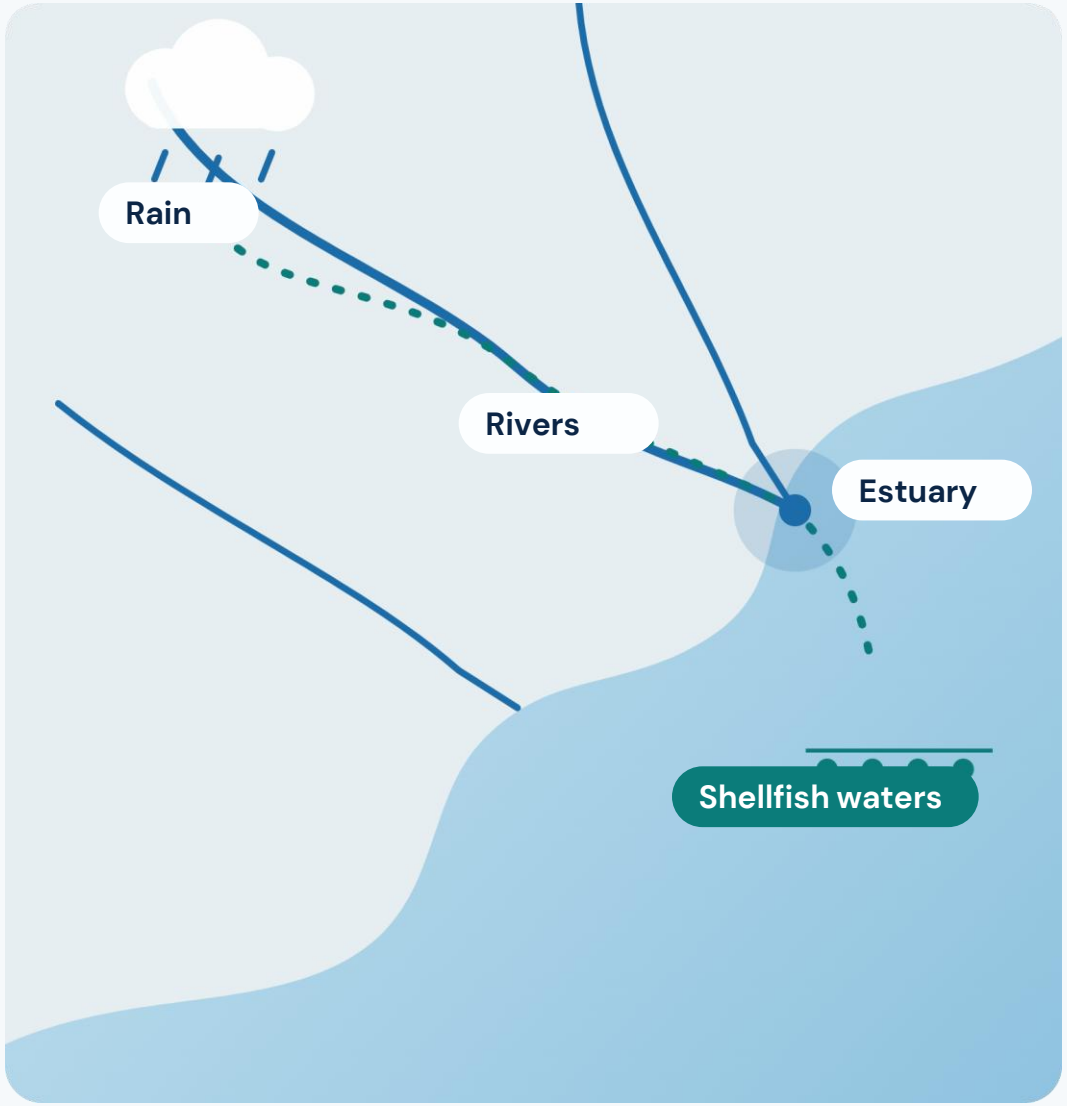
RunOffPulse Ireland

AI-Driven Coastal Runoff Early Warning & Risk Attribution System

ORK01: COASTAL RUNOFF WATCH

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WHY IT MATTERS

Every heavy rainfall/runOff sends the land into the sea



Runoff carries faecal bacteria, nutrients and sediment to the coast within hours to days.

A PERSONAL CASE

Illustrative scenario

Meet Sam, a mussel grower in Cork Harbour

SATURDAY

20 mm of rain falls in two days.

SUNDAY

Rivers across the Lee, Bandon and Blackwater rise. Nobody tells her.

MONDAY

She harvests as normal. The sample result arrives days later.

> **The warning signs were in public data all along, just scattered across four agencies.**

The Unseen Coastal Blindspot

By the time a water sample reaches the lab, the damage is already done.

Ireland's marine economy relies on **100% manual testing**—leaving aquaculture farms and coastal councils blind to toxic agricultural runoff spikes until 48 hours after impact.

Our Paradigm Shift
Transforming reactive 48-hour lag into 24-hour predictive intelligence.

48h
Detection Lag
Lab culture delay during critical runoff events.

€35M+
Annual Industry Loss
Unpredicted shellfish harvest & beach closures.

100%
Legacy Vulnerability
Over-reliance on manual shoreline sampling.

WHO SHOULD CARE

Who is affected

Shellfish & aquaculture operators
Product quality, harvest planning

Local authorities & councils
Pollution investigation, beach management

Environmental consultancies
Catchment and coastal studies

Coastal communities, swimmers & water-sports groups
Safe use of the sea

THE PROBLEM TODAY

The data exists. The joined-up picture doesn't.

Met Éireann

Rainfall

OPW

Live river levels

EPA

Catchments, estuary health ratings, shellfish waters, monitoring stations

beaches.ie

EPA beach bacteria tests

No one combines them into a single early warning for the coast.

OUR SOLUTION

RunoffPulse: one live view from rain to sea

RunoffPulse
IRELAND

- Overview
- Coastal Risk Map
- Beach Forecast (AI)
- Shellfish Watch
- Water Quality
- Rain & Rivers
- Community Reports
- Data Sources
- Business Value

Cork coast pilot. Early-warning support only, not an official water-quality or shellfish advisory.

From rainfall to coastal risk, before it becomes a bigger problem.

API connected

Coastal water status

Rainfall, rivers and coastal waters around Cork Harbour, Kinsale and Youghal, combined to show where land runoff may be reaching the sea.

Now | Rain event · 27 Sep 2026 | Storm Babet · 18 Oct 2023

CORK HARBOUR & COAST · NOW

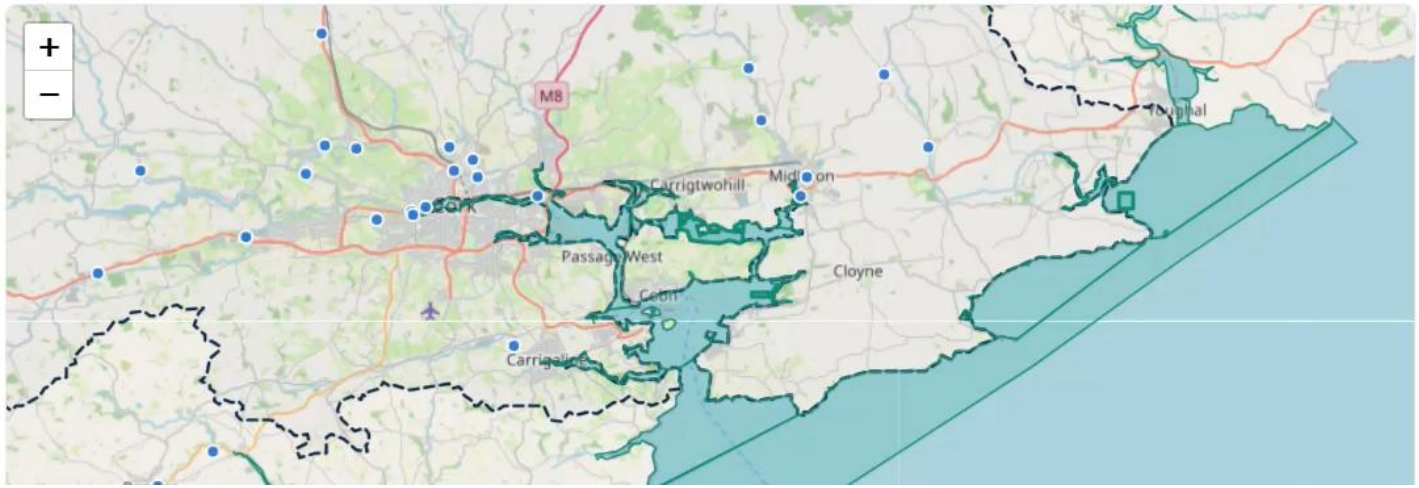
Normal conditions

Little rain (0.3 mm in 2 days), rivers are at normal levels, and none of the 34 coastal areas need extra attention.

Open coastal map →

Rain, last 2 days	Rivers running high	Coastal areas needing attention	Beach water forecast (AI)
0.3 mm	0 of 62	0 of 34	16%
Met Éireann, Cork Airport	OPW river gauges above their usual 30-day range	Estuaries, bays and shellfish waters	Highest chance of a failed bacteria test: Garryvoe (off-season estimate)

Coastal areas and rivers



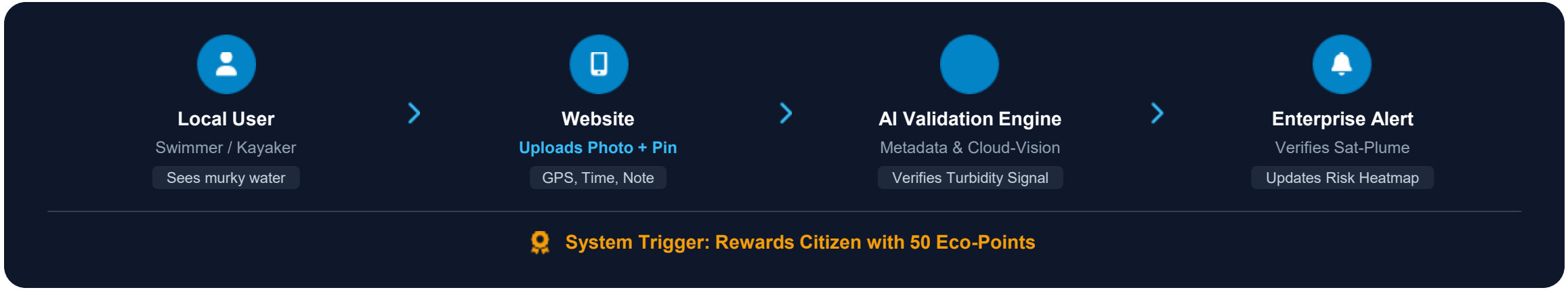
Coastal areas

All coastal areas are normal. Examples:

Ballycotton Bay Coastal water	Normal
Ballymacoda Bay Shellfish water	Normal
Cork Great Island North Channel Shellfish water	Normal
Cork Harbour Coastal water	Normal
Cuskinny Lake Estuary	Normal
Glashaboy Estuary Estuary	Normal

INCENTIVIZED CITIZEN SCIENCE

How the Eco-Points System Works



1 Spot & Pin

Kayaking in Cork Harbour or swimming at Myrtleville, users snap & geotag algal blooms or muddy runoff via app.

2 Metadata & Check

App automatically embeds GPS, time, weather, and tide data while image AI validates visual water turbidity.

3 Ground-Truthing

Backend cross-references user telemetry with Copernicus Sentinel-3 and Met Éireann feeds to confirm hotspot.

4 Reward Credited

Upon automated validation, user profile receives +50 Eco-Points instantly to boost community engagement.

PROOF IT WORKS

Tested on real events and 12 years of real data

Real event replay · 27 Sep 2026

20 mm of rain in 48 hours

30 of 62 river gauges high

34 of 34 coastal areas flagged High

AI Beach Forecast

13% vs 2%

of tests failed on days rated ‘Raised chance’ vs ‘Low chance’, on years the AI had never seen

Trained on 1,714 real EPA beach tests, 2015–2026, 24 Cork beaches

Real but modest signal. Built to prioritise checks, not replace lab tests.

WHAT'S NEW AND DIFFERENT

What's already out there, and where we fit

Solution	What it gives you	The gap
beaches.ie (EPA)	Bathing water results after sampling	Not predictive
SFPA / Marine Institute	Official shellfish classifications	Reported after testing
waterlevel.ie (OPW)	Live river levels	No link to the coast
catchments.ie (EPA)	Catchment science and assessments	Not real-time
Safer Seas & Rivers (UK)	Sewage overflow alerts	UK only
Sensor buoys & instruments	Precise measurements	Costly, single-point
RunoffPulse	Rain, rivers and coast in one view	Adds the early warning

We don't replace official testing. We add the missing early warning between rain and results, using data that's free and already collected.

MAIN ADVANTAGE

Our Defensive Moat

Predictive vs. Reactive

24–48h Advance Risk Alerts: While legacy tools report yesterday's water quality, our engine uses precipitation triggers and satellite streaming to predict runoff risks before contamination hits the coast.

Software-First, Zero-CAPEX

Open Data + Eco-Points: Replaces multi-million euro buoy hardware with Copernicus satellite telemetry, Met Éireann feeds, and citizen verification from local swimmers/kayakers.

Dual B2G + B2B Commercial Model

Economic Value Loop: Saves councils money on emergency lab sampling (B2G) while protecting shellfish harvest yields and avoiding unannounced beach closures (B2B).

COMMERCIAL & MONETIZATION MODEL

Dual B2G & B2B Monetization Framework

Multi-Tiered Revenue

RunoffPulse Coastal Risk Engine

Core Intelligence Infrastructure • Continuous Satellite Data + Predictive AI + IoT Sensing

Real-Time Risk Scoring

24-Hour Predictive Lead

PUBLIC SECTOR

B2G: Municipalities & Authorities

Stakeholders: Cork County Council, EPA Ireland, Inland Fisheries

Cost Avoidance & Efficiency
Direct cleanup cost reduction + zero emergency sampling fees (€500/sample).

Regulatory Risk Shield
Prevents EU compliance non-compliance fines & Blue Flag beach revocations.

Annual Enterprise SaaS License
€15k – €40k / county / yr

PRIVATE SECTOR

B2B: Marine & Coastal Enterprise

Stakeholders: Aquaculture & Oysters, Coastal Resorts, Water Sports

Yield & Stock Protection
Prevents mass shellfish mortality from sudden agricultural runoff & microbial spikes.

Brand Upside & Trust
Premium eco-certification + live "Clean Water Guarantee" widgets for guests.

Tiered Subscriptions & API Access
€99 – €1,000 / mo

ROADMAP & CLOSE

Next: from Cork Harbour to every Irish coast

NOW

Cork Harbour, Kinsale, Youghal: 3 catchments, 34 coastal areas

NEXT

Connect SFPA shellfish classifications, Marine Institute biotoxin data, tide data

THEN

Satellite water colour and turbidity (Sentinel), radar rainfall, all Irish coasts

The sea tells us about the land, a few days too late. RunoffPulse tells you first.

Live demo: rain event replay, 27 Sep 2026

RunoffPulse

Thank You.